

and vitamins and minerals. There is now significant evidence to show that chicory root protects the liver from damage.

History & Folklore According to Pliny (23–79 CE), chicory juice was mixed with rose oil and vinegar as a remedy for headaches. The roasted root is commonly used as a coffee substitute. The young root can be boiled and eaten like parsnips.

Medicinal Actions & Uses Chicory is an excellent mild bitter tonic for the liver and digestive tract. The root is therapeutically similar to dandelion root (*Taraxacum officinale*, p. 141), supporting the action of the stomach and liver and cleansing the urinary tract. Chicory is also taken for rheumatic conditions and gout, and as a mild laxative, one particularly appropriate for children. An infusion of the leaves and flowers also aids digestion.

Related Species The endive (*C. endiva*) has similar though milder effects.

Cinnamomum camphora syn. *Laurus camphora* (Lauraceae) Camphor

Description Evergreen tree growing to 100 ft (30 m). Produces red leaves that turn dark green as they mature, small fragrant yellowish flowers, and oval red berries.

Habitat & Cultivation Native to China and Japan, this tree is cultivated in tropical and sub-tropical regions for its wood, from which camphor oil is derived.

Parts Used Stems, root, wood, leaves, twigs, volatile oil.

Constituents The plant contains a volatile oil comprising camphor, saffrole, eugenol, and terpineol. It also contains lignans. Camphor is irritant and antiseptic; saffrole is thought to be carcinogenic. A white crystalline substance derived from the stems, root, and other parts of the tree, also called camphor, has powerful antiseptic, stimulant, and antispasmodic properties.

History & Folklore In the 13th century, Marco Polo noted that camphor oil was highly valued by the Chinese as a medicine, scent, and embalming fluid.

Medicinal Actions & Uses Camphor is most commonly applied externally as a counterirritant and analgesic liniment to relieve arthritic and rheumatic pains, neuralgia, and back pain. It may be applied to skin problems, such as cold sores and chilblains, and used as a chest rub for bronchitis and other chest infections. Though the oil has been taken for various complaints, internal use is not advised.

Related Species See also cinnamon (*Cinnamomum* spp., p. 81).

Cautions Do not take internally. Camphor oil is subject to legal restrictions in some countries.

Citrullus vulgaris (Cucurbitaceae) Watermelon

Description Annual vine with hairy, 3–5 lobed leaves, yellow flowers, and green fruit typically growing to 10 in (25 cm) across.

Habitat & Cultivation Native to tropical Africa, watermelon is grown throughout warm temperate to tropical regions. The fruit is gathered when ripe.

Parts Used Fruit, seeds.

Constituents Watermelon contains citrullin and arginine, both of which are thought to increase urea production in the liver, so increasing the flow of urine.

History & Folklore Watermelon species have been used in Egypt for more than 4,000 years, figuring in wall paintings dating to the Old Kingdom (2686–2181 BCE). They appear to have been a component in remedies for trembling fingers, constipation, and expelling disease brought on by demons. Egyptian myth recounts that the watermelon originated from the semen of the god Seth.

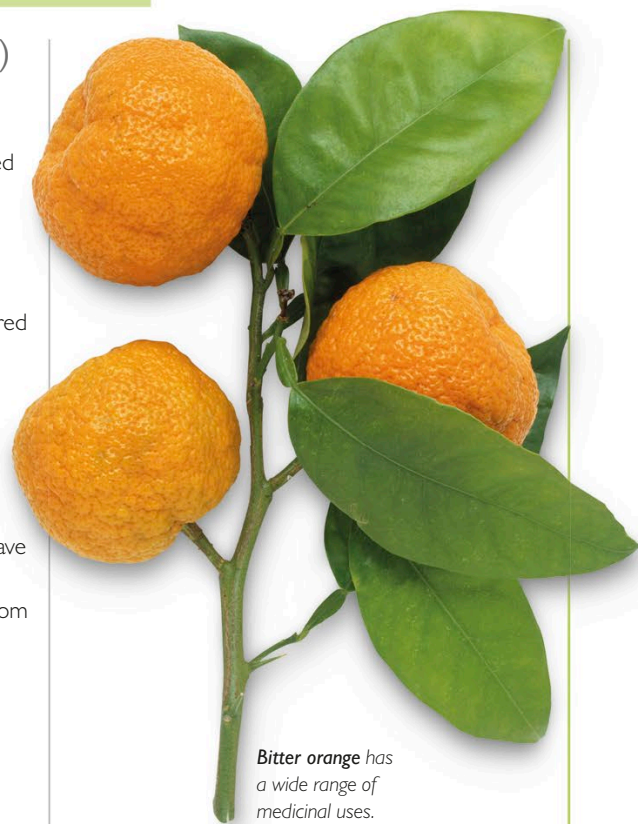
Medicinal Actions & Uses Watermelon is best known as a thirst-quenching fruit that comes into season when temperatures are at their hottest. In traditional Chinese medicine it is used precisely to counter “summer heat” patterns—characterized by excessive sweating, thirst, raised temperature, scanty urine, diarrhea, and irritability or anger. Watermelon fruit and juice soothe these symptoms, increasing urine flow and cleansing the kidneys. The fruit’s refreshing properties extend to the digestive system, where it clears gas. Watermelon may also be used in the treatment of hepatitis. In hot, stifling weather it is helpful for those suffering from bronchitis or asthma. The cooling fruit pulp may be applied to hot and inflamed skin and to soothe sunburn. The seeds can be mashed and used to expel worms.

Related Species The watermelons of Egypt (*C. lanata* and *C. colocynthoides*) are very similar species. The colocynth (*C. colocynthus*), native to dry areas of Africa and Asia, is extremely bitter and contains a cucurbitacin glycoside with antitumor properties.

Citrus aurantium (Rutaceae) Bitter Orange

Description Evergreen tree growing to 30 ft (10 m). Has leathery, dark green leaves, delicately perfumed white flowers, and orange fruit.

Habitat & Cultivation Native to tropical Asia, this tree is now grown throughout the tropics and subtropics. Orchards of bitter orange are found along the Mediterranean coast, especially in Spain.



Bitter orange has a wide range of medicinal uses.

Parts Used Fruit, peel, leaves, flowers, seeds, essential oil.

Constituents Bitter orange peel contains a volatile oil with limonene (around 90%), flavonoids, coumarins, triterpenes, vitamin C, carotene, and pectin. The flavonoids are anti-inflammatory, antibacterial, and antifungal. The composition of the volatile oils in the leaves, flowers and peel varies significantly. Linalyl acetate (50%) is the main constituent in oil from the leaves (petitgrain) and linalool (35%) in oil from the flowers (neroli). The unripe fruit of the bitter orange contains citranin, which reportedly is contraceptive.

History & Folklore The bitter orange has provided food and medicine for thousands of years. It yields neroli oil from its flowers, and the oil known as petitgrain from its leaves and young shoots. Both distillates are used extensively in perfumery. Orange flower water is a by-product of distillation and is used in perfumery and to flavor candies and cookies, as well as being used medicinally.

Medicinal Actions & Uses The strongly acidic fruit of the bitter orange stimulates digestion and relieves flatulence. An infusion of the fruit is thought to soothe headaches, calm palpitations, and lower fevers. The juice helps the body eliminate waste products and, being rich in vitamin C, helps the immune system ward off infection. If taken to excess, however, its acid content can exacerbate arthritis. In Chinese herbal medicine, the unripe fruit, known as *zhi shi*, is thought to “regulate the qi,” helping to relieve flatulence and abdominal bloating, and to open the bowels. The